

ESM4. Classification Reports

Tables I to X present partial classification reports for selected configurations under **Protocol A**, including the best-performing model (DenseNet201 + Fine-tuned + Attention) and the best baseline model (ResNet50 + Frozen + No Attention). We additionally include Table XI, which presents a partial classification report for the best-performing model under **Protocol B** (MobileNetV3-Large + Fine-tuned + Attention). Across all tables, we **bold** specific rows to emphasize key results, with ellipses (...) indicating that additional rows exist before and after the displayed subset.

TABLE I: Classification Report for DenseNet201 + Fine-tuned + No Attention (Protocol A)

Class Name	Mean (Std)			
	Precision	Recall	F1-Score	Support
⋮	⋮	⋮	⋮	⋮
Yousef Hobeiche	1.0000 (0.0000)	0.9949 (0.0089)	0.9974 (0.0045)	65.0000 (0.0000)
Angele Ellis	0.9744 (0.0444)	1.0000 (0.0000)	0.9867 (0.0231)	12.0000 (0.0000)
Yousef Hobeiche & Angele Ellis	0.6667 (0.5774)	0.6667 (0.5774)	0.6667 (0.5774)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
Elias Abdallah	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	3.0000 (0.0000)
Youssef Bey Al-Helou	1.0000 (0.0000)	0.8333 (0.2887)	0.8889 (0.1925)	2.0000 (0.0000)
Father Elias	0.6667 (0.5774)	0.6667 (0.5774)	0.6667 (0.5774)	1.0000 (0.0000)
Magistrate of Batroun	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
accuracy	0.9862 (0.0035)	0.9862 (0.0035)	0.9862 (0.0035)	0.9862 (0.0035)
macro avg	0.9638 (0.0205)	0.9561 (0.0215)	0.9557 (0.0226)	2849.0000 (0.0000)
weighted avg	0.9863 (0.0041)	0.9862 (0.0035)	0.9852 (0.0041)	2849.0000 (0.0000)

TABLE II: Classification Report for the Best Model (DenseNet201 + Fine-tuned + Attention (Protocol A)

Class Name	Mean (Std)			
	Precision	Recall	F1-Score	Support
⋮	⋮	⋮	⋮	⋮
Yousef Hobeiche	1.0000 (0.0000)	0.9949 (0.0089)	0.9974 (0.0045)	65.0000 (0.0000)
Angele Ellis	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	12.0000 (0.0000)
Yousef Hobeiche & Angele Ellis	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
Elias Abdallah	0.9167 (0.1443)	1.0000 (0.0000)	0.9524 (0.0825)	3.0000 (0.0000)
Youssef Bey Al-Helou	1.0000 (0.0000)	0.8333 (0.2887)	0.8889 (0.1925)	2.0000 (0.0000)
Father Elias	0.3333 (0.5774)	0.3333 (0.5774)	0.3333 (0.5774)	1.0000 (0.0000)
Magistrate of Batroun	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
accuracy	0.9905 (0.0009)	0.9905 (0.0009)	0.9905 (0.0009)	0.9905 (0.0009)
macro avg	0.9819 (0.0039)	0.9730 (0.0040)	0.9744 (0.0028)	2849.0000 (0.0000)
weighted avg	0.9912 (0.0006)	0.9905 (0.0009)	0.9902 (0.0007)	2849.0000 (0.0000)

TABLE III: Classification Report for the Best Baseline Model (ResNet50 + Frozen + No Attention) (Protocol A)

Class Name	Mean (Std)			
	Precision	Recall	F1-Score	Support
⋮	⋮	⋮	⋮	⋮
Yousef Hobeiche	0.8490 (0.0410)	0.9128 (0.0622)	0.8787 (0.0355)	65.0000 (0.0000)
Angele Ellis	0.5541 (0.0495)	0.8889 (0.1273)	0.6799 (0.0565)	12.0000 (0.0000)
Yousef Hobeiche & Angele Ellis	0.1667 (0.2887)	0.1667 (0.2887)	0.1667 (0.2887)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
Elias Abdallah	0.3333 (0.5774)	0.1111 (0.1925)	0.1667 (0.2887)	3.0000 (0.0000)
Youssef Bey Al-Helou	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	2.0000 (0.0000)
Father Elias	0.3333 (0.5774)	0.3333 (0.5774)	0.3333 (0.5774)	1.0000 (0.0000)
Magistrate of Batroun	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
accuracy	0.8652 (0.0035)	0.8652 (0.0035)	0.8652 (0.0035)	0.8652 (0.0035)
macro avg	0.7387 (0.0190)	0.6648 (0.0168)	0.6772 (0.0162)	2849.0000 (0.0000)
weighted avg	0.8701 (0.0079)	0.8652 (0.0035)	0.8572 (0.0052)	2849.0000 (0.0000)

The classification report for the best baseline model shows that the model struggles with classes having few samples, yielding very low precision, recall, and F1-scores. For example, "Youssef Bey Al-Helou," "Father Elias," "Magistrate of Batroun," and "Elias Abdallah" suffer from limited samples and sparse training representation. This data scarcity likely prevents the model from learning meaningful writer-specific patterns, resulting in poor generalization and misidentification during evaluation.

TABLE IV: Classification Report for ResNet50 + Frozen + Attention (Protocol A)

Class Name	Mean (Std)			
	Precision	Recall	F1-Score	Support
⋮	⋮	⋮	⋮	⋮
Yousef Hobeiche	0.9393 (0.0266)	0.9538 (0.0533)	0.9462 (0.0363)	65.0000 (0.0000)
Angele Ellis	0.7075 (0.0417)	1.0000 (0.0000)	0.8282 (0.0286)	12.0000 (0.0000)
Yousef Hobeiche & Angele Ellis	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
Elias Abdallah	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	3.0000 (0.0000)
Youssef Bey Al-Helou	0.3333 (0.5774)	0.3333 (0.5774)	0.3333 (0.5774)	2.0000 (0.0000)
Father Elias	0.3333 (0.5774)	0.3333 (0.5774)	0.3333 (0.5774)	1.0000 (0.0000)
Magistrate of Batroun	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
accuracy	0.9114 (0.0050)	0.9114 (0.0050)	0.9114 (0.0050)	0.9114 (0.0050)
macro avg	0.8146 (0.0225)	0.7792 (0.0171)	0.7795 (0.0169)	2849.0000 (0.0000)
weighted avg	0.9135 (0.0066)	0.9114 (0.0050)	0.9069 (0.0043)	2849.0000 (0.0000)

TABLE V: Classification Report for DenseNet201 Fine-tuned + Last 25 Layers + No Attention (Protocol A)

Class Name	Mean (Std)			
	Precision	Recall	F1-Score	Support
⋮	⋮	⋮	⋮	⋮
Yousef Hobeiche	0.9800 (0.0228)	0.9846 (0.0154)	0.9821 (0.0115)	65.0000 (0.0000)
Angele Ellis	0.8654 (0.0449)	0.8889 (0.0481)	0.8767 (0.0418)	12.0000 (0.0000)
Yousef Hobeiche & Angele Ellis	0.6667 (0.5774)	0.3333 (0.2887)	0.4444 (0.3849)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
Elias Abdallah	1.0000 (0.0000)	0.4444 (0.1925)	0.6000 (0.1732)	3.0000 (0.0000)
Youssef Bey Al-Helou	0.6667 (0.5774)	0.6667 (0.5774)	0.6667 (0.5774)	2.0000 (0.0000)
Father Elias	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	1.0000 (0.0000)
Magistrate of Batroun	0.3333 (0.5774)	0.3333 (0.5774)	0.3333 (0.5774)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
accuracy	0.9404 (0.0093)	0.9404 (0.0093)	0.9404 (0.0093)	0.9404 (0.0093)
macro avg	0.8659 (0.0165)	0.8362 (0.0293)	0.8353 (0.0230)	2849.0000 (0.0000)
weighted avg	0.9430 (0.0085)	0.9404 (0.0093)	0.9375 (0.0096)	2849.0000 (0.0000)

TABLE VI: Classification Report for DenseNet201 Fine-tuned + Last 25 Layers + Attention (Protocol A)

Class Name	Mean (Std)			
	Precision	Recall	F1-Score	Support
⋮	⋮	⋮	⋮	⋮
Yousef Hobeiche	0.9750 (0.0228)	0.9897 (0.0089)	0.9823 (0.0158)	65.0000 (0.0000)
Angele Ellis	0.7951 (0.0776)	0.9167 (0.0833)	0.8470 (0.0287)	12.0000 (0.0000)
Yousef Hobeiche & Angele Ellis	0.5000 (0.5000)	0.3333 (0.2887)	0.3889 (0.3469)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
Elias Abdallah	0.8000 (0.3464)	0.6667 (0.0000)	0.7000 (0.1732)	3.0000 (0.0000)
Youssef Bey Al-Helou	0.6667 (0.3333)	0.6667 (0.2887)	0.6222 (0.2037)	2.0000 (0.0000)
Father Elias	0.3333 (0.5774)	0.3333 (0.5774)	0.3333 (0.5774)	1.0000 (0.0000)
Magistrate of Batroun	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
accuracy	0.9479 (0.0112)	0.9479 (0.0112)	0.9479 (0.0112)	0.9479 (0.0112)
macro avg	0.8897 (0.0237)	0.8703 (0.0266)	0.8659 (0.0272)	2849.0000 (0.0000)
weighted avg	0.9520 (0.0106)	0.9479 (0.0112)	0.9462 (0.0116)	2849.0000 (0.0000)

TABLE VII: Classification Report for DenseNet201 + Frozen + No Attention (Protocol A)

Class Name	Mean (Std)			
	Precision	Recall	F1-Score	Support
⋮	⋮	⋮	⋮	⋮
Yousef Hobeiche	0.9233 (0.0281)	0.9795 (0.0178)	0.9503 (0.0150)	65.0000 (0.0000)
Angele Ellis	0.6825 (0.1689)	0.7222 (0.1925)	0.7009 (0.1769)	12.0000 (0.0000)
Yousef Hobeiche & Angele Ellis	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
Elias Abdallah	1.0000 (0.0000)	0.5556 (0.1925)	0.7000 (0.1732)	3.0000 (0.0000)
Youssef Bey Al-Helou	0.3333 (0.5774)	0.3333 (0.5774)	0.3333 (0.5774)	2.0000 (0.0000)
Father Elias	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	1.0000 (0.0000)
Magistrate of Batroun	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
accuracy	0.8705 (0.0043)	0.8705 (0.0043)	0.8705 (0.0043)	0.8705 (0.0043)
macro avg	0.6346 (0.1136)	0.6179 (0.1043)	0.6092 (0.1060)	2849.0000 (0.0000)
weighted avg	0.8660 (0.0062)	0.8705 (0.0043)	0.8644 (0.0064)	2849.0000 (0.0000)

TABLE VIII: Classification Report for DenseNet201 + Frozen + Attention (Protocol A)

Class Name	Mean (Std)			
	Precision	Recall	F1-Score	Support
⋮	⋮	⋮	⋮	⋮
Yousef Hobeiche	0.9919 (0.0045)	0.9851 (0.0076)	0.9884 (0.0033)	65.0000 (0.0000)
Angele Ellis	0.9111 (0.0244)	0.9444 (0.0824)	0.9222 (0.0434)	12.0000 (0.0000)
Yousef Hobeiche & Angele Ellis	1.0000 (0.0000)	0.6667 (0.2887)	0.7778 (0.1925)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
Elias Abdallah	0.9167 (0.1443)	0.7778 (0.1925)	0.8190 (0.0330)	3.0000 (0.0000)
Youssef Bey Al-Helou	0.8889 (0.1925)	0.6667 (0.2887)	0.7111 (0.0770)	2.0000 (0.0000)
Father Elias	0.3333 (0.5774)	0.3333 (0.5774)	0.3333 (0.5774)	1.0000 (0.0000)
Magistrate of Batroun	1.0000 (0.0000)	0.6667 (0.2887)	0.7778 (0.1925)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
accuracy	0.9686 (0.0041)	0.9686 (0.0041)	0.9686 (0.0041)	0.9686 (0.0041)
macro avg	0.9251 (0.0297)	0.8920 (0.0364)	0.9013 (0.0327)	2849.0000 (0.0000)
weighted avg	0.9683 (0.0044)	0.9686 (0.0041)	0.9662 (0.0049)	2849.0000 (0.0000)

TABLE IX: Classification Report for DenseNet201 + From Scratch + No Attention (Protocol A)

Class Name	Mean (Std)			
	Precision	Recall	F1-Score	Support
⋮	⋮	⋮	⋮	⋮
Yousef Hobeiche	0.9850 (0.0149)	0.9949 (0.0089)	0.9898 (0.0043)	65.0000 (0.0000)
Angele Ellis	0.8990 (0.0364)	0.9722 (0.0481)	0.9332 (0.0234)	12.0000 (0.0000)
Yousef Hobeiche & Angele Ellis	0.6667 (0.5774)	0.5000 (0.5000)	0.5556 (0.5092)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
Elias Abdallah	1.0000 (0.0000)	0.6667 (0.3333)	0.7667 (0.2517)	3.0000 (0.0000)
Youssef Bey Al-Helou	0.6667 (0.5774)	0.5000 (0.5000)	0.5556 (0.5092)	2.0000 (0.0000)
Father Elias	0.3333 (0.5774)	0.3333 (0.5774)	0.3333 (0.5774)	1.0000 (0.0000)
Magistrate of Batroun	1.0000 (0.0000)	0.6667 (0.2887)	0.7778 (0.1925)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
accuracy	0.9711 (0.0016)	0.9711 (0.0016)	0.9711 (0.0016)	0.9711 (0.0016)
macro avg	0.9242 (0.0147)	0.9042 (0.0149)	0.9052 (0.0152)	2849.0000 (0.0000)
weighted avg	0.9702 (0.0033)	0.9711 (0.0016)	0.9687 (0.0026)	2849.0000 (0.0000)

TABLE X: Classification Report for DenseNet201 + From Scratch + Attention (Protocol A)

Class Name	Mean (Std)			
	Precision	Recall	F1-Score	Support
⋮	⋮	⋮	⋮	⋮
Yousef Hobeiche	0.9949 (0.0087)	0.9897 (0.0089)	0.9923 (0.0001)	65.0000 (0.0000)
Angele Ellis	0.9209 (0.0037)	0.9722 (0.0481)	0.9456 (0.0250)	12.0000 (0.0000)
Yousef Hobeiche & Angele Ellis	1.0000 (0.0000)	0.6667 (0.2887)	0.7778 (0.1925)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
Elias Abdallah	0.9167 (0.1443)	0.7778 (0.1925)	0.8190 (0.0330)	3.0000 (0.0000)
Youssef Bey Al-Helou	0.8889 (0.1925)	0.6667 (0.2887)	0.7111 (0.0770)	2.0000 (0.0000)
Father Elias	0.3333 (0.5774)	0.3333 (0.5774)	0.3333 (0.5774)	1.0000 (0.0000)
Magistrate of Batroun	1.0000 (0.0000)	0.6667 (0.2887)	0.7778 (0.1925)	2.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
accuracy	0.9674 (0.0046)	0.9674 (0.0046)	0.9674 (0.0046)	0.9674 (0.0046)
macro avg	0.9298 (0.0153)	0.9030 (0.0283)	0.9060 (0.0240)	2849.0000 (0.0000)
weighted avg	0.9679 (0.0049)	0.9674 (0.0046)	0.9653 (0.0052)	2849.0000 (0.0000)

TABLE XI: Classification Report for MobileNetV3-Large + Fine-tuned + Attention (Protocol B)

Class Name	Mean (Std)			
	Precision	Recall	F1-Score	Support
⋮	⋮	⋮	⋮	⋮
Emily Nasrallah	1.0000 (0.0000)	1.0000 (0.0000)	1.0000 (0.0000)	11.0000 (0.0000)
Jean Philipp Kmeid	0.9982 (0.0025)	1.0000 (0.0000)	0.9991 (0.0013)	27.0000 (0.0000)
Hanna Moussa	1.0000 (0.0000)	0.9978 (0.0031)	0.9989 (0.0016)	25.0000 (0.0000)
Salah Tizani	0.9876 (0.0131)	1.0000 (0.0000)	0.9937 (0.0067)	297.0000 (0.0000)
Hanna Ghayth (Juan Gait)	0.8343 (0.0182)	1.0000 (0.0000)	0.9096 (0.0108)	13.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
Botros Feghaly	0.9583 (0.0589)	0.9702 (0.0212)	0.9626 (0.0207)	24.0000 (0.0000)
Elie Maroun Khalil	0.9247 (0.0140)	1.0000 (0.0000)	0.9608 (0.0075)	22.0000 (0.0000)
Murshid Habashi	0.8674 (0.1030)	0.9720 (0.0200)	0.9139 (0.0639)	26.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
Father Botros Hasan	0.2428 (0.1358)	0.6667 (0.4423)	0.3416 (0.2150)	19.0000 (0.0000)
Father Youssef Hanna & Father Botros Hasan	0.2529 (0.3576)	0.3188 (0.4509)	0.2821 (0.3989)	23.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
May Ziadeh	0.2480 (0.1763)	0.6515 (0.4611)	0.3591 (0.2549)	22.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
Prof. Ign. Kratchkovsky	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	13.0000 (0.0000)
Asaad Koury Tobia Attallah	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	10.0000 (0.0000)
Amin Farhat	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	21.0000 (0.0000)
⋮	⋮	⋮	⋮	⋮
accuracy	0.7707 (0.0111)	0.7707 (0.0111)	0.7707 (0.0111)	0.7788 (0.0000)
macro avg	0.6460 (0.0065)	0.6865 (0.0135)	0.6317 (0.0141)	2473.0000 (0.0000)
weighted avg	0.7712 (0.0180)	0.7707 (0.0111)	0.7468 (0.0136)	2473.0000 (0.0000)